GLOBAL ENERGY TRANSITION

& Market Reliability

Data Visualisation Project Report

United Kingdom · United States · Australia

Coverage: 2019–2025 (Australia: 1999–2025)

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1. Introduction

The global electricity sector is undergoing its most significant structural transformation since electrification. Solar and wind generation capacity are growing at unprecedented rates, yet their intermittent nature raises fundamental questions about grid reliability and market stability. This project investigates those questions through a comparative analysis of three national electricity grids — the United Kingdom, the United States, and Australia — which represent three distinct stages and approaches to the energy transition.

The United Kingdom has pursued a deliberate coal phase-out, completing it in September 2024, while expanding offshore wind to cover over 30% of annual generation. The United States operates the world's largest electricity market with significant regional diversity, transitioning more slowly due to structural inertia and political variation. Australia, heavily dependent on coal historically, has experienced one of the fastest renewable buildouts globally, driven by exceptional solar resources and competitive market incentives.

Three research questions guide the investigation:

- **RQ1 — Transition Velocity:** Between 2019 and 2025, which country moved away from fossil fuels (Coal + Gas) fastest? The speed of fossil decline is compared against Wind and Solar growth across all three grids to identify the most aggressive transitioner.
- **RQ2 — Grid Stress-Testing:** During peak demand periods — winter nights in the UK, summer heatwaves in Australia, and summer heat in the US — when renewable output is low, which backup sources fill the gap, and how fossil-dependent does each grid become under stress?

- **RQ3 — Market Volatility:** Does a higher share of variable renewables (Wind + Solar) make wholesale electricity prices more volatile? Statistical measures (Standard Deviation, Coefficient of Variation) are applied to the Australian wholesale market — the only dataset with price data — spanning 1999 to 2025.
- **RQ4 — Portfolio Diversification:** Is the energy transition making electricity grids structurally more diverse and resilient, or simply replacing one dominant fuel with another? The Herfindahl-Hirschman Index (HHI) — a concentration metric from economics — is applied to each country's monthly generation mix to quantify portfolio diversity over time.

The motivation for this work is practical: policymakers, investors, and grid operators need evidence-based answers to these questions as they design transition pathways. The analysis combines three publicly available datasets, processed through a four-stage R pipeline, and visualised using interactive Shiny dashboards.

2. Data Wrangling and Checking

2.1 Data Sources

Three national-level datasets were obtained from official grid operators and energy statistics agencies:

Dataset	Country	Provider	URL	Granularity	Coverage
gridwatch.csv	UK	GridWatch / National Grid ESO	gridwatch.templar.co.uk	5-minute	2019–2025
US Energy generation dataset.csv	USA	EIA Hourly Grid Monitor	eia.gov/electricity/gridmonitor	Daily	2019–2025
19990101 All Regions Australia.csv	AU	AEMO / OpenNEM	opennem.org.au	Monthly	1999–2026

Raw file sizes were substantial: the UK GridWatch file contained 734,336 rows, the US EIA production file 5,048,813 rows, and the Australian AEMO file 326 rows. Together, these represented over 5.7 million records before processing.

2.2 Data Wrangling Pipeline

A four-script R pipeline was developed using tidyverse, lubridate, and janitor. Each script handles one country and outputs a standardised monthly CSV; the fourth script combines all three.

```
Pipeline: 01_clean_uk.R → 02_clean_us.R → 03_clean_australia.R → 04_combine.R
Output:   uk_monthly.csv | us_monthly.csv | au_monthly.csv → combined_energy.csv
```

United Kingdom — GridWatch

- **Leading whitespace:** All 734,336 timestamp values contained a leading space character (' 2019-01-01 00:00:36'). This caused silent NA production in all datetime parsers until str_trim() was applied before ymd_hms().
- **Data gaps:** 13 gaps exceeding one hour were identified by computing consecutive timestamp differences. The largest gap was 72.3 hours in March 2025. These were documented as known limitations rather than imputed, as monthly aggregation reduces their impact to below 0.5%.
- **Sensor dropouts:** 2,888 rows recorded frequency = 0.000 Hz — physically impossible on the UK grid (nominal 50 Hz). These rows were retained since generation data remained valid; the frequency column was excluded from all analysis.

- **Gas technology merger:** The raw data splits gas into ccgt (Combined Cycle, baseload) and ocgt (Open Cycle, peaking). Both were recoded to 'Gas' after pivot_longer() to match the single NG code used by the US EIA.
- **Unit conversion:** Each 5-minute MW instantaneous reading was converted to GWh using: GWh = MW × (5 ÷ 60) ÷ 1,000, then summed monthly.

United States — EIA

- **Scale:** The production file contained 5,048,813 rows — five times the development sample of 1,055,000 rows.
- **readr type coercion bug:** The period column ('2019-01-01' format) was silently coerced to a Date object by readr before mutate() was called, causing ymd_h() and parse_date_time() to fail with 'All formats failed to parse.' Fix: col_types = cols(period = col_character()) at read time, plus as.character() as a defensive cast.
- **Duplicate rows:** 4,029,404 rows (79.8%) were duplicates arising from overlapping utility-level and pre-aggregated regional records. These were collapsed by group_by(country, date, source) |> summarise().
- **Negative values:** 151,492 rows carried negative generation values representing battery charging, pumped hydro charging, and grid imports. These were removed with filter(value >= 0).
- **Unmapped fuel codes:** 7 new EIA codes (BAT, SNB, PS, OES, WNB, UNK, UES) representing hybrid storage technologies were absent from the original fuel_map dictionary. These were mapped to appropriate categories (Other, Solar, Hydro, Wind).
- **Unit conversion:** EIA values are in MWh. Divided by 1,000 to produce GWh, then summed monthly.

Australia — AEMO / OpenNEM

- **Structural zeros:** Solar (zero before 2016), Wind (zero before 2009), and Bioenergy (zero before 2009) carried zero values in early years — not missing data, but correct records of zero generation before deployment. Retained as valid data points.
- **Negative generation:** 6 rows held negative generation_gwh from pumped storage charging periods. Removed in the combine step with filter(generation_gwh >= 0).
- **Price missing marker:** The 'Volume Weighted Price - AUD/MWh' column used 0 as a missing-value marker in years with sparse reporting. Zero values were replaced with NA before any price analysis.
- **No unit conversion required:** The AEMO file is already monthly GWh.

2.3 Cross-Country Harmonisation

- **Units:** All three datasets were standardised to GWh per month.
- **Source vocabulary:** 12 standardised source labels were defined. CCGT + OCGT → Gas. Pumped storage → Hydro. EIA codes mapped to readable names.
- **Renewable flag:** A boolean renewable column was added: TRUE for Wind, Solar, Hydro, Bioenergy, and Biomass.

- **Scale disparity:** The US generates ~260× more electricity than the UK or Australia. All cross-country comparisons therefore use percentage share of total generation, not absolute GWh.
- **Date filter:** The combined dataset uses 2019–2025 as the common window. Australia's full 1999–2025 data is retained separately for RQ3 price analysis.

2.4 Data Quality Verification

Check	UK	US	AU	Combined
Null values	**0**	**0**	**0**	**0**
Negative GWh (post-filter)	**0**	**0**	**0**	**0**
Date range correct	**Yes**	**Yes**	**Yes**	**Yes**
Expected row count	**504**	**686**	**2,268**	**1,778**
All sources present	**6**	**9**	**7**	**12**
Renewable flag coverage	N/A	N/A	N/A	**1,778/1,778**

All checks passed after the wrangling pipeline. The combined dataset is fully clean with zero nulls, no negative values, correct date coverage, and consistent source vocabulary across all three countries.

3. Data Exploration

The exploration was conducted in three phases, one per research question, using R Shiny dashboards with plotly visualisations. The choice of interactive dashboards was deliberate: electricity data is highly seasonal and multi-dimensional, and static charts cannot adequately support the filtering and drill-down needed to examine patterns across countries, years, and source types simultaneously. All tools used were R, plotly, bslib, lubridate, tidyverse, and the slider package for rolling windows.

3.1 RQ1 — Transition Velocity

Objective and design rationale

> Question: Between 2019 and 2025, which country moved away from fossil fuels (Coal + Gas) fastest, and which grew Wind + Solar the quickest?

- **What we are looking for:** The rate of change in the generation mix — not just the current level, but how fast each country is moving. A country at 60% fossil declining by 0.3 pp/month is transitioning faster than one at 40% declining by 0.1 pp/month, even though the latter has a cleaner grid today.
- **How we measure it:** Monthly percentage share of Coal + Gas ('fossil share') and Wind + Solar ('W+S share') are computed for each country. Linear regression (lm() in R) is fitted to each time series, producing a slope coefficient in percentage points per month — the 'transition velocity.' This single number is directly comparable across countries regardless of grid size.
- **Why percentage share, not absolute GWh:** The US generates ~260× more electricity than the UK or Australia. Without normalising to percentages, the US dominates every absolute chart and meaningful comparison is impossible.
- **Why line charts with trend overlay:** Time series line charts (Figures 1–4) show both the raw monthly data and the fitted regression line. The raw line reveals seasonal volatility; the dashed trend line isolates the underlying direction. Bar charts cannot show continuous temporal trends; scatter plots would lose the time axis.
- **Why a slope comparison bar chart:** Figure 5 translates the regression slopes into a grouped bar chart — enabling direct side-by-side comparison of a single scalar metric (pp/month) across countries. The zero reference line distinguishes decline from growth.
- **Why a dumbbell chart:** The dumbbell plot (in dashboard) shows start-year and end-year values as connected dots. It answers 'how far did each country travel?' rather than 'how fast?' — complementing the slope chart with absolute magnitude.

Approach and justification

Percentage share of monthly generation was chosen as the primary metric rather than absolute GWh, because the US generates approximately 260 times more electricity than the UK. Without normalisation, visual comparison is impossible. Linear regression was applied to each country's monthly fossil share and Wind+Solar share time series, producing a slope coefficient (percentage points per month) as the 'transition velocity' — a single comparable number per country.

Visual analysis — fossil share decline

Figure 1 plots the monthly fossil (Coal + Gas) share of generation for all three countries from January 2019 to December 2025. The raw monthly lines reveal the volatility that trend lines alone would hide: Australia's share fluctuates seasonally around a clear downward trajectory, the UK exhibits sharp month-to-month swings driven by wind variability, and the US moves within a tight band around 55–65%, confirming its structural inertia.

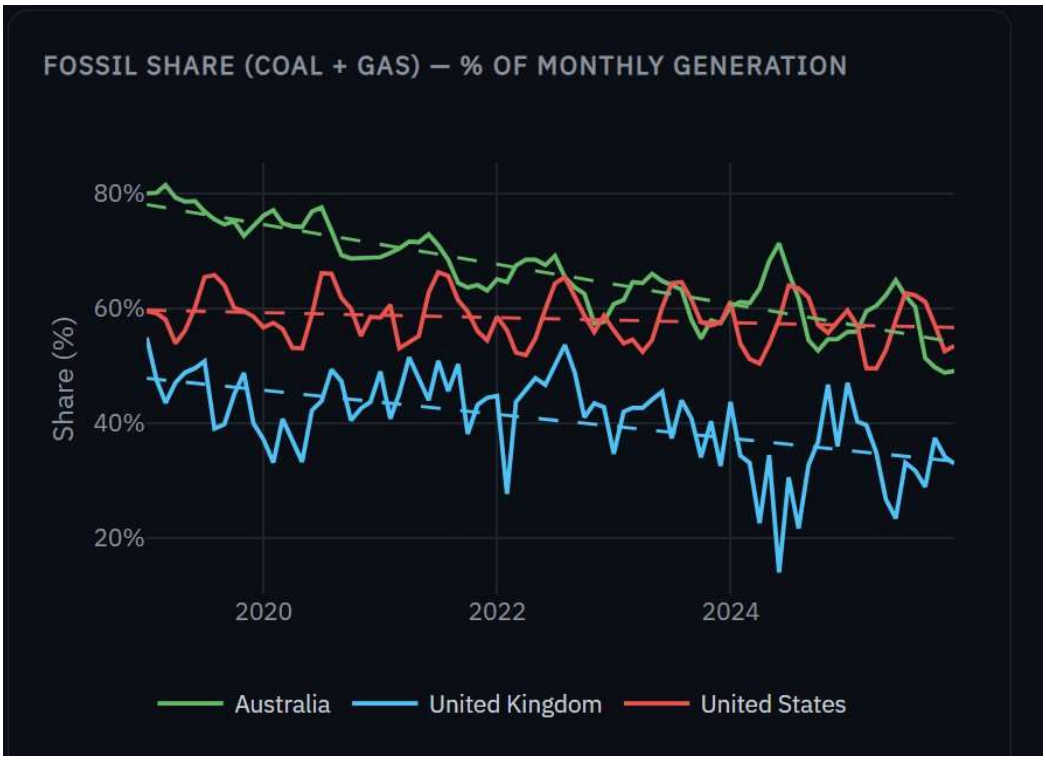

Figure 1. Monthly fossil share (Coal + Gas) across UK, US, and Australia, 2019–2025. Dashed lines show linear regression fits. All three countries show declining trends, but Australia's slope is visibly steepest.

Figure 2 isolates the linear regression fits from Figure 1, making the relative speeds of decline visually unambiguous. Australia's trajectory crosses the US line in 2024 — despite starting ~17 percentage points above it in 2019. The UK sits on a parallel lower path, reflecting its earlier coal exit.

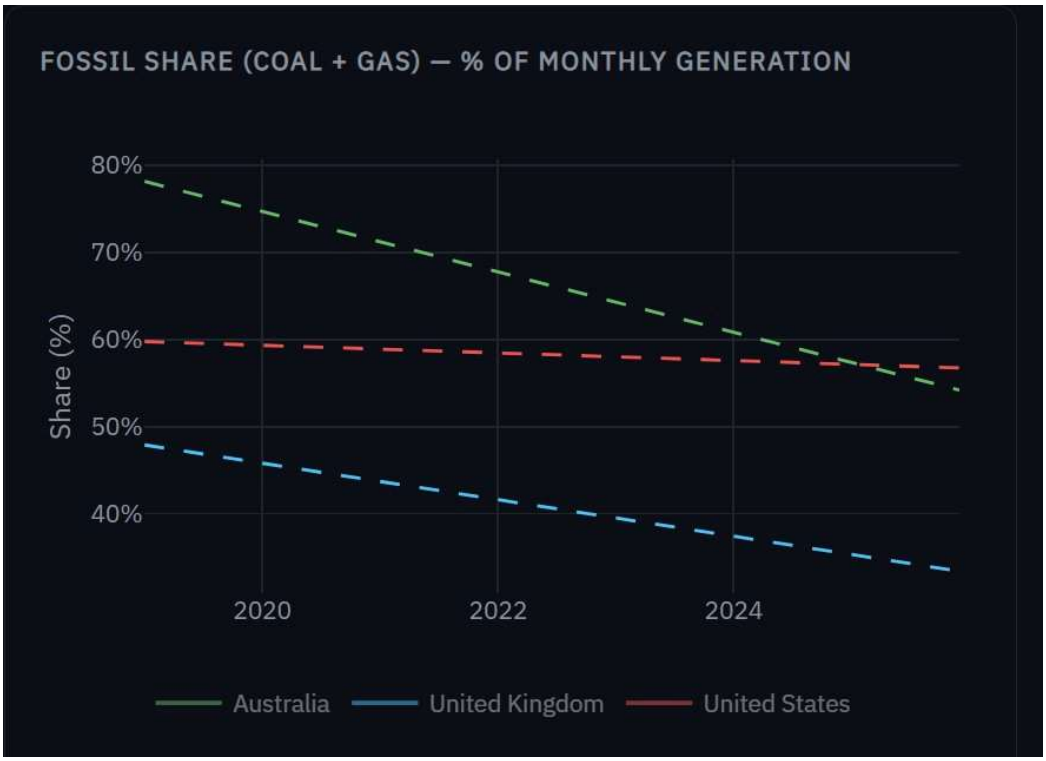

Figure 2. Linear regression fits only — fossil share trend lines for UK, US, and Australia, 2019–2025. Australia (green) shows the steepest decline, crossing below the US (red) by 2024.

Visual analysis — Wind + Solar growth

Figure 3 shows the corresponding trend lines for Wind + Solar share. The three paths are nearly parallel in slope — but at very different levels. The UK and Australia both approach 40% Wind+Solar by 2025, while the US reaches only 18%. Critically, Australia's slope matches the UK's despite starting 7 percentage points lower.

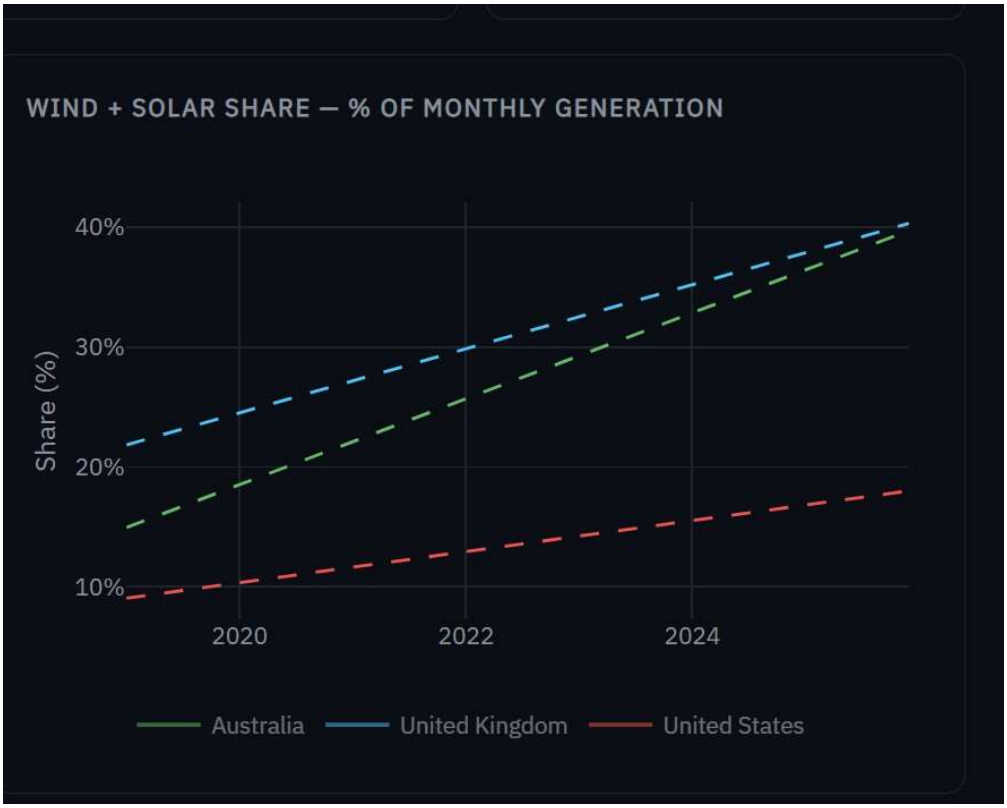

Figure 3. Wind + Solar share — trend lines only. UK and Australia follow near-parallel growth paths reaching ~40% by 2025; the US trails at ~18%.

Figure 4 overlays the raw monthly data on the trend lines. The UK's extreme month-to-month variability — spikes above 50% in some months, dips below 20% in others — demonstrates wind intermittency. Australia's seasonal pattern is more regular, reflecting the stable solar resource base.

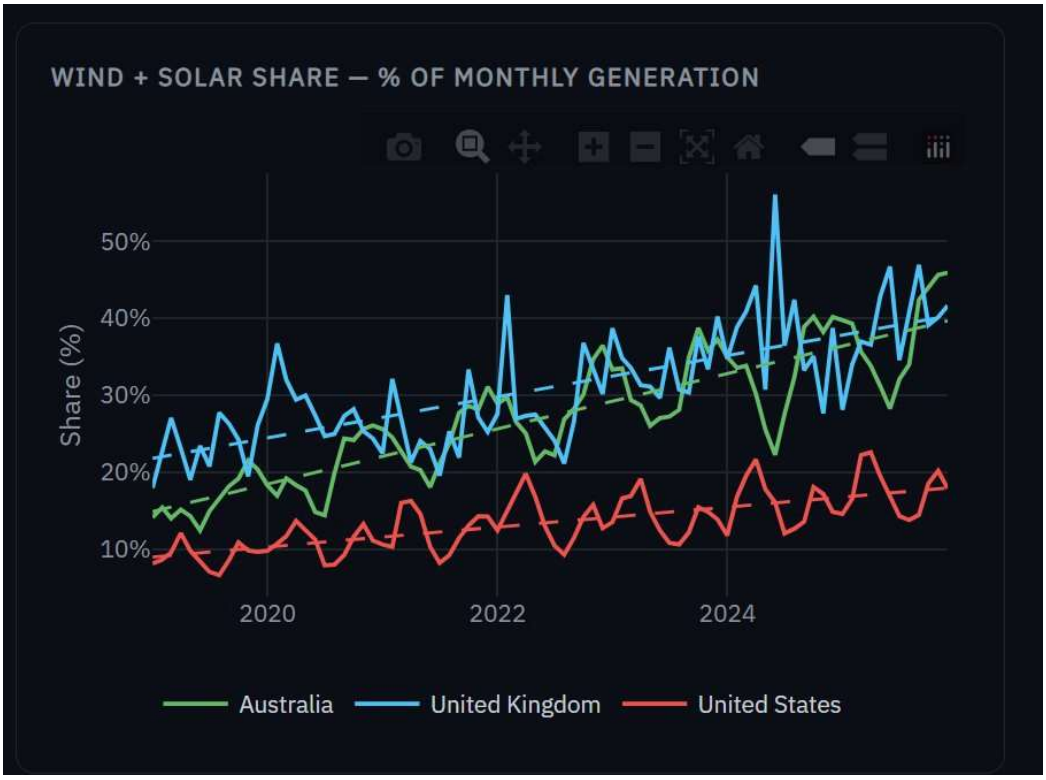

Figure 4. Wind + Solar share with raw monthly data. UK shows extreme month-to-month volatility reflecting wind intermittency; Australia exhibits a smoother seasonal pattern driven by solar.

Visual analysis — transition velocity comparison

Figure 5 distils the regression results into a single direct comparison. The bar heights are the slope coefficients from the linear regressions in Figures 1–4. Australia's fossil decline (−0.289 pp/month) is roughly 1.7× faster than the UK's and 8× faster than the US's. Australia also has the strongest Wind+Solar growth (+0.298 pp/month). The symmetry between the bars within each country is striking — the fossil energy being displaced is being replaced almost one-for-one by variable renewables.

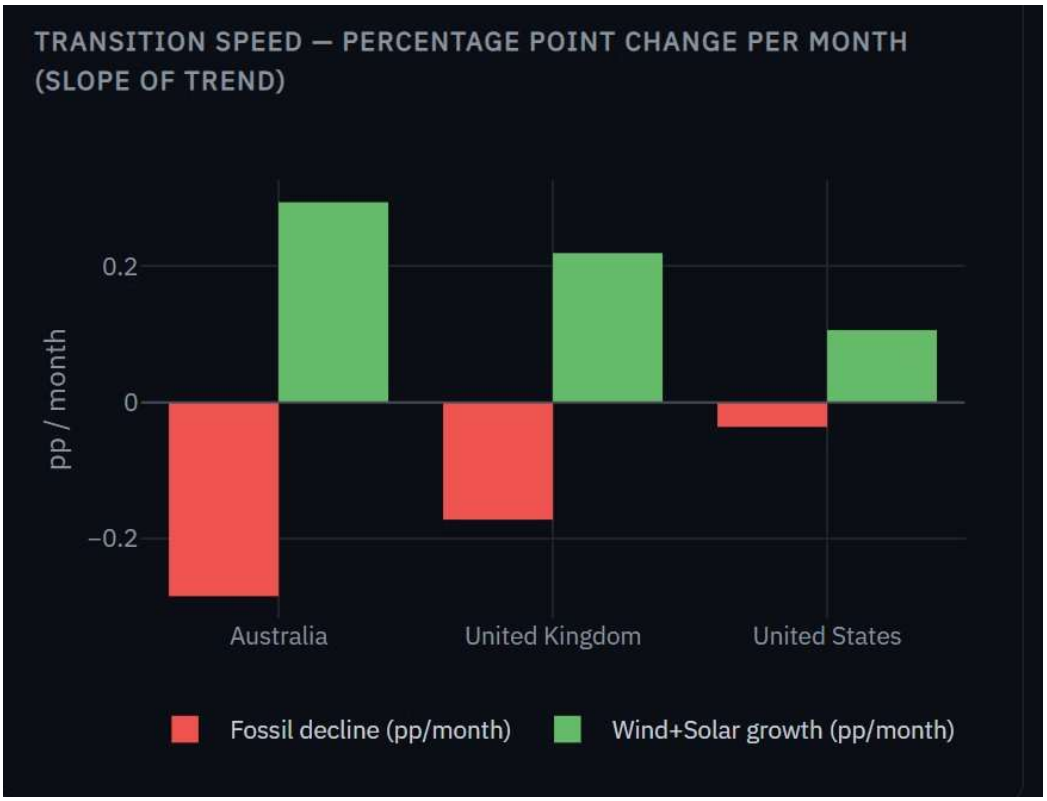

Figure 5. Transition speed comparison — linear regression slope (percentage points per month) for fossil decline (red) and Wind+Solar growth (green). Australia is the fastest transitioner on both metrics.

Key findings

Country	Fossil 2019	Fossil 2025	Δ Fossil	Slope (pp/mo)	W+S 2019	W+S 2025	Δ W+S	Slope (pp/mo)
Australia	77.3%	56.8%	**-20.5 pp**	**-0.289**	16.4%	37.7%	**+21.4 pp**	**+0.298**
United Kingdom	46.3%	34.2%	-12.1 pp	-0.175	23.2%	39.1%	+15.9 pp	+0.223
United States	60.1%	56.4%	**-3.7 pp**	**-0.036**	9.2%	17.7%	+8.5 pp	+0.108

- **Australia fastest transitioner:** Australia recorded the steepest fossil decline slope (−0.289 pp/month) and the largest absolute shift of any country (−20.5 pp fossil, +21.4 pp Wind+Solar). Crucially, Australia started from the highest fossil base (77.3% in 2019), demonstrating that even a deeply coal-dependent grid can transition rapidly with the right policy and resource conditions.
- **UK steady and structural:** The UK's transition predates the study window — it entered 2019 already at 46.3% fossil, having exited coal years earlier. Its slope (−0.175 pp/month) reflects a mature, ongoing transition rather than a sudden shift.
- **US structural inertia:** The US moved slowest on both metrics (−0.036 pp/month fossil decline). The scale and diversity of the US grid — 81 utility respondents, large coal-dependent regions — create structural barriers that aggregate statistics mask.
- **Coal absence in UK:** Coal does not appear in the UK processed dataset — a real-world finding, not a data gap. Great Britain's last coal plant closed in September 2024.

Visualisations used for RQ1

- **Line charts (fossil share & Wind+Solar share):** Monthly time series for all three countries on shared axes with trend lines overlaid. Line charts were chosen over bar charts because the continuous temporal trend is the key quantity of interest. Figures 1–4 demonstrate the value of showing raw and fitted data side-by-side.
- **Slope comparison bar chart (Figure 5):** A grouped bar chart comparing fossil_slope and ws_slope across countries. This translates the regression outputs into a directly scannable visual showing which country changed fastest on each metric.
- **Dumbbell chart (in dashboard):** Start (2019) and end (2025) fossil share plotted as dots connected by a horizontal line per country. This makes the absolute magnitude of change immediately readable, which line charts do not emphasise.
- **Stacked area chart (in dashboard):** Full generation mix by source over time, faceted by country. Useful for confirming that observed share changes reflect real source shifts, not merely total generation fluctuations.

3.2 RQ2 — Grid Stress-Testing

Objective and design rationale

> Question: During periods of peak demand when renewable output is low, how does each grid survive? Which backup sources fill the gap?

- **What we are looking for:** The 'reliability gap' — how much the renewable share of generation drops during peak-stress months compared to normal months. A large gap means the grid becomes significantly more fossil-dependent under stress, revealing structural vulnerability.
- **How we define peak stress:** Each country has a different climate-driven peak: UK winters (Dec–Feb) when solar is near zero and heating demand peaks; Australian summers (Dec–Feb) when extreme heat drives air-conditioning load; US summers (Jun–Aug) when heat waves stress southern and central grids.
- **How we measure it:** Renewable share is computed separately for peak and non-peak months, then the difference (non-peak minus peak) gives the reliability gap in percentage points. The backup source mix is computed as each source's share of total generation during peak months only.
- **Why donut charts for backup mix:** Figure 6 uses donut charts because the question is 'what fraction of the whole does each source contribute?' — a part-of-whole composition question that donuts make instantly readable. Stacked bars would work but are harder to compare when the totals differ.
- **Why a gap line chart:** Figure 7 plots the reliability gap annually as a line chart. A line was chosen because the key insight is the trend — is the gap widening or narrowing? The zero reference line provides the interpretive threshold.
- **Why a scatter with peak highlighting:** In the dashboard, monthly dots are plotted with peak months as large bright markers and non-peak as faded dots. This reveals seasonal patterns and provides granular evidence for the aggregate gap statistic.

Approach and justification

Peak stress periods were defined per country based on known grid stress patterns: UK winters (December–February) when solar output is near zero and heating demand peaks; Australian summers (December–February) when extreme heat drives air-conditioning load to record highs; and US summers (June–August) when heat events stress the southern and central grids. The reliability gap was computed as the difference in renewable share between non-peak and peak months — a positive gap means the grid's clean energy share falls during stress periods, revealing fossil backup dependency.

Visual analysis — backup source mix during peak

Figure 6 presents three donut charts side-by-side showing what sources each grid relies on during its peak-stress season. The composition differences are dramatic and reveal each country's structural strategy.

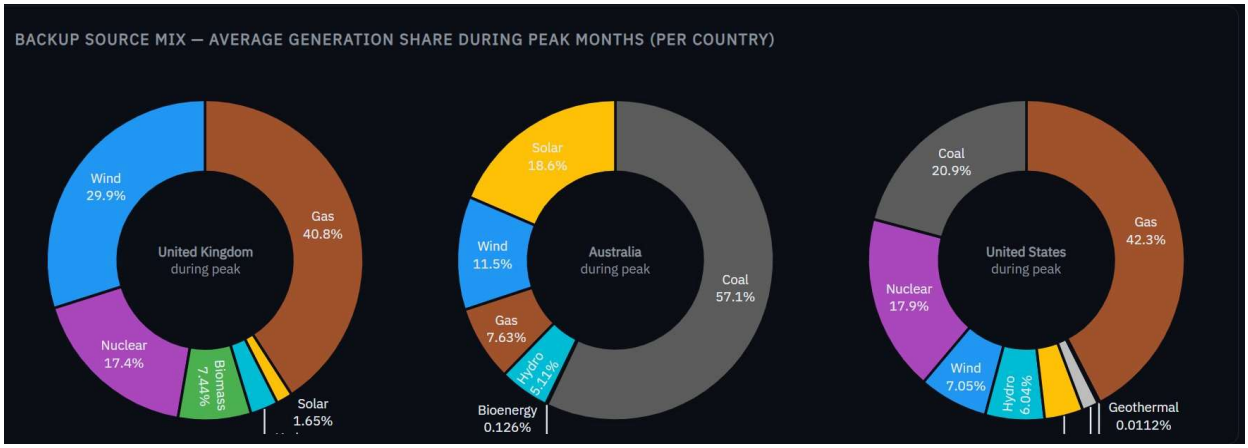

Figure 6. Backup source mix during peak months — UK (winter), Australia (summer), and US (summer). Donut charts show each country's dispatchable backup dependency.

Three observations emerge from Figure 6. First, the UK's peak mix is the most diversified — Gas (40.8%), Wind (29.9%), Nuclear (17.4%), and Biomass (7.4%) form a balanced portfolio where no single source dominates. Second, Australia's peak generation is 57.1% Coal alone — the legacy coal fleet still carries the grid during summer stress, even as Solar contributes 18.6%. Third, the US depends on three fossil sources plus Nuclear, with Gas (42.3%) and Coal (20.9%) together covering nearly two-thirds of peak demand.

Visual analysis — reliability gap over time

Figure 7 plots the reliability gap (non-peak renewable share minus peak renewable share) annually for each country. A positive gap means renewable share drops during peak months; a negative gap means renewable share is actually higher during peak. The orange dashed line at zero is the critical reference point.

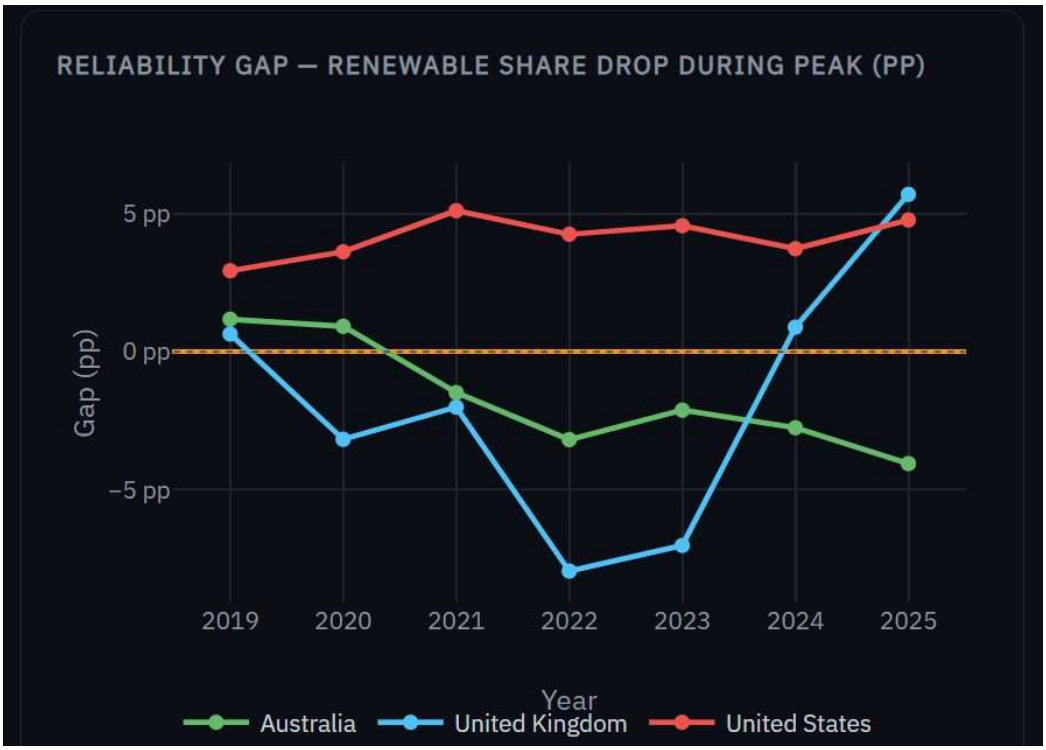

Figure 7. Reliability gap over time — renewable share drop during peak (percentage points). The US consistently runs positive (gap grows during summer); UK and Australia hover around zero or negative.

The US (red line) is consistently above zero across the entire 2019–2025 window — its renewable share always drops during summer peaks because solar output cannot compensate for collapsed wind generation during heatwaves. Australia (green) stays below zero most years: its solar resource is strongest precisely when demand peaks, so renewable share is actually higher in summer than winter. The UK (blue) shows the most volatile behaviour, with sharp dips to −8 pp in 2022–2023 (extreme winter wind shortages) and a jump to +6 pp in 2025.

Visual analysis — fossil dependency trend

Figure 8 shows how fossil share during peak months has evolved year-by-year. Where the reliability gap chart above shows the relative change, this chart shows the absolute fossil dependency level during stress periods.

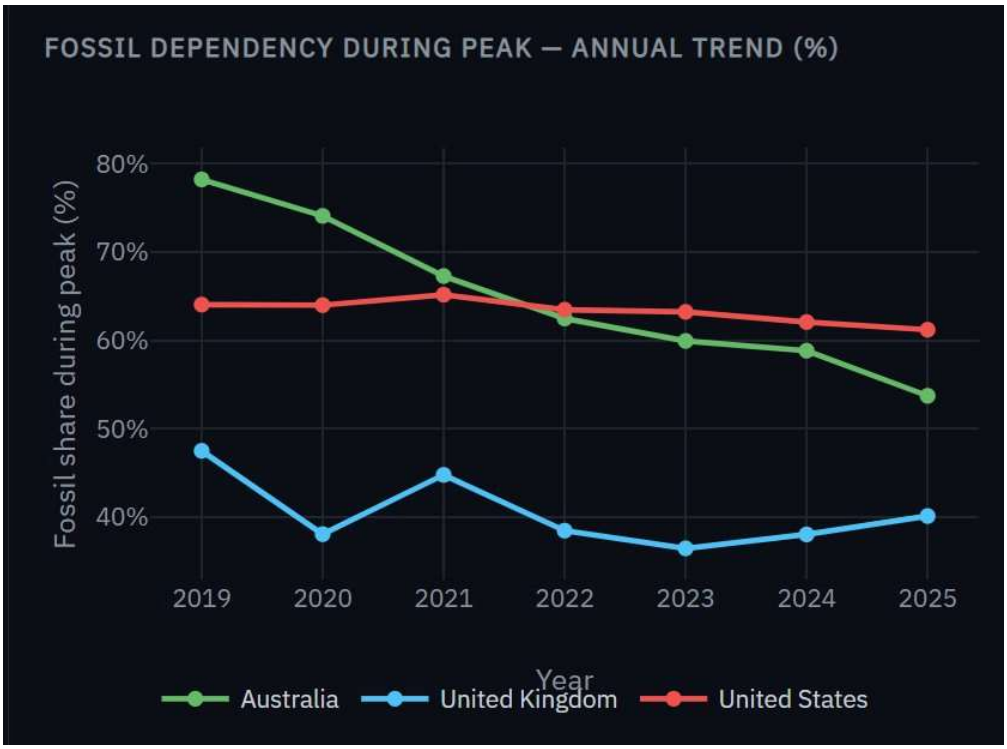

Figure 8. Fossil dependency during peak — annual trend. Australia's peak fossil share fell sharply from 78.2% (2019) to 53.7% (2025); UK stayed below 50%; US declined slowly from 64% to 61%.

Australia's trajectory is the most striking — peak fossil dependency fell from 78.2% in 2019 to 53.7% in 2025, a 24.5 percentage point reduction in just six years. The UK stayed consistently below 50% throughout, reflecting its mature transition. The US declined only marginally from 64% to 61%, confirming that its peak-period fossil lock-in is substantially stronger than the other two grids.

Key findings

Country	Peak season	Renewable (peak)	Renewable (non-peak)	Reliability gap	Primary backup
United Kingdom	Winter (Dec–Feb)	41.8%	39.9%	-1.9 pp	Nuclear + Gas
Australia	Summer (Dec–Feb)	35.2%	33.5%	-1.7 pp	Coal + Gas
United States	Summer (Jun–Aug)	**17.0%**	**21.1%**	**-4.1 pp**	Gas + Coal + Nuclear

- **US most vulnerable:** The US exhibits the largest reliability gap (−4.1 pp). During summer peaks, renewable share drops from 21.1% to 17.0% while fossil dependency rises to 63.4%. Gas (42.3%) and Coal (20.9%) together cover nearly two-thirds of generation during peak stress (Figure 6).
- **UK most resilient:** The UK's reliability gap is the smallest (−1.9 pp). Nuclear (17.4% of peak generation) provides stable baseload unaffected by weather, and Wind is often strong in winter months — meaning peak and non-peak renewable shares are nearly equal.
- **Australia coal-heavy backup:** Despite significant renewable growth, Coal still dominates peak-period generation in Australia (57.1% of peak mix — Figure 6). This reveals the structural persistence of coal as the primary backup even as Solar and Wind expand rapidly.
- **Annual trend — improving resilience:** All three countries show declining peak-period fossil dependency year-over-year from 2019 to 2025 (Figure 8). Australia's peak fossil share fell from 78.2% to 53.7%, suggesting that growing renewable capacity is gradually reducing stress-period vulnerability even before storage solutions are deployed at scale.

Visualisations used for RQ2

- **Donut charts (Figure 6):** Backup source mix during peak months shown as donut/pie charts, one per country. Chosen over bars because the composition-of-a-whole question is naturally suited to a part-whole chart format. Side-by-side placement enables direct comparison of mix structure.
- **Reliability gap line chart (Figure 7):** The difference (non-peak minus peak renewable share) plotted annually per country. A line chart was chosen to show whether the gap is widening or narrowing over time, which a single bar cannot capture. Zero reference line provides the critical interpretive threshold.
- **Fossil dependency line chart (Figure 8):** Peak-period fossil share tracked annually. Provides the absolute dependency level that complements the relative reliability gap chart.
- **Grouped bar chart (in dashboard):** Renewable share during peak and non-peak months shown side by side per country. Simple, direct comparison of the reliability gap.
- **Monthly scatter with peak highlighted (in dashboard):** Individual monthly data points coloured by peak status. Provides granular evidence for the aggregate statistics and reveals seasonal patterns within years.
- **Stacked bar — annual backup breakdown (in dashboard):** Year-by-year source breakdown during peak months, faceted by country. Allows tracking of how the backup mix evolves — for example, whether Gas is gradually replacing Coal in Australia's peak mix.

3.3 RQ3 — Market Volatility

Objective and design rationale

> Question: Does a higher share of Solar and Wind make wholesale electricity prices more volatile and unpredictable? (Australia only — only dataset with price data)

- **What we are looking for:** Whether renewable penetration is statistically associated with price instability. The hypothesis to test: 'more variable renewables = more price swings.' If true, the scatter plot should show an upward trend; if false, volatility is driven by other factors.
- **How we measure volatility:** Two metrics are used. Standard Deviation (SD) measures absolute price swings in AUD/MWh. Coefficient of Variation (CV = SD ÷ Mean × 100) measures relative volatility — essential because average prices tripled over the study period, so a rising SD alone could reflect higher price levels, not more instability.
- **Why three historical eras:** The 26-year dataset is divided into Pre-renewable (1999–2009, W+S ≈0%), Early transition (2010–2016, W+S ≈5%), and High renewable (2017–2025, W+S ≈25%). Comparing distributions across eras tests whether the regime change coincides with higher volatility.
- **Why a rolling 12-month SD:** Figure 9 uses a rolling window rather than annual aggregates because it captures when volatility regimes shift without being dominated by calendar-year boundaries.
- **Why a scatter plot for the core test:** The W+S share vs. CV scatter is the critical analytical chart. The scatter idiom was chosen because the hypothesis concerns the relationship between two continuous variables — no other chart type can test this directly.
- **Why box plots for era comparison:** Box plots show the full price distribution within each era (median, IQR, outliers) — far more informative than comparing era means alone.
- **Why an exclude-2022 toggle:** 2022 is a confounding outlier driven by the global energy crisis. An interactive toggle lets the analyst test whether the correlation survives without this single dominant event — it does not, weakening the hypothesis significantly.

Approach and justification

Price data is only available for Australia (AEMO monthly wholesale price, AUD/MWh, 1999–2025). Two volatility metrics were computed: Standard Deviation (SD) measures absolute price swings, while Coefficient of Variation (CV = SD ÷ Mean × 100) measures relative volatility — essential because nominal Australian electricity prices tripled over the study period, meaning rising SD alone could simply reflect higher prices rather than more instability. Annual CV was correlated with annual Wind+Solar share to test the core hypothesis. Three historical eras were defined to contextualise the analysis: Pre-renewable (1999–2009), Early transition (2010–2016), and High renewable (2017–2025).

Visual analysis — rolling price volatility

Figure 9 shows the rolling 12-month standard deviation of Australian wholesale electricity prices from 1999 to 2025. This metric acts as a seismograph for market instability — each spike corresponds to a discrete event that destabilised wholesale prices. The pre-2015 period is remarkably flat, with rolling SD rarely exceeding $20 AUD/MWh. Three major spikes punctuate the record: 2017 (South Australia gas crisis), 2019 (drought-driven hydro shortage), and the dominant 2022 peak reaching $120 AUD/MWh — the largest volatility event on record, driven by coal plant outages and global gas price contagion from the Ukraine conflict.

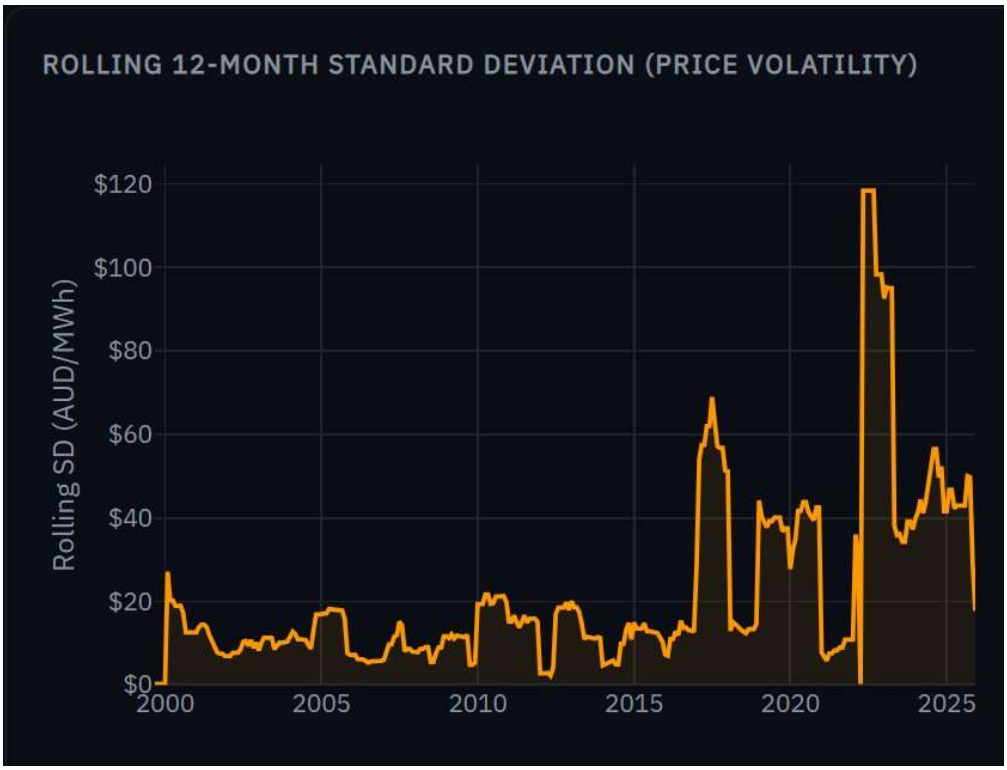

Figure 9. Rolling 12-month standard deviation of Australian wholesale electricity prices (AUD/MWh), 1999–2025. Major spikes in 2017 and 2022 correspond to fossil fuel supply shocks, not renewable intermittency events.

Visual analysis — annual coefficient of variation

Figure 10 presents the annual CV coloured by era — grey for the pre-renewable period (1999–2009), orange for early transition (2010–2016), and blue for the high-renewable era (2017–2025). If renewables systematically caused volatility, the blue bars should be consistently taller than the orange and grey bars. Instead, the pattern is irregular: 2020 (CV = 69%, driven by COVID demand collapse) and 2022 (CV = 57%, energy crisis) are the highest blue bars, but 2010 (CV = 60%, carbon tax uncertainty) in the orange era is comparable. Several blue-era years (2018, 2021, 2025) have lower CV than the pre-renewable baseline.

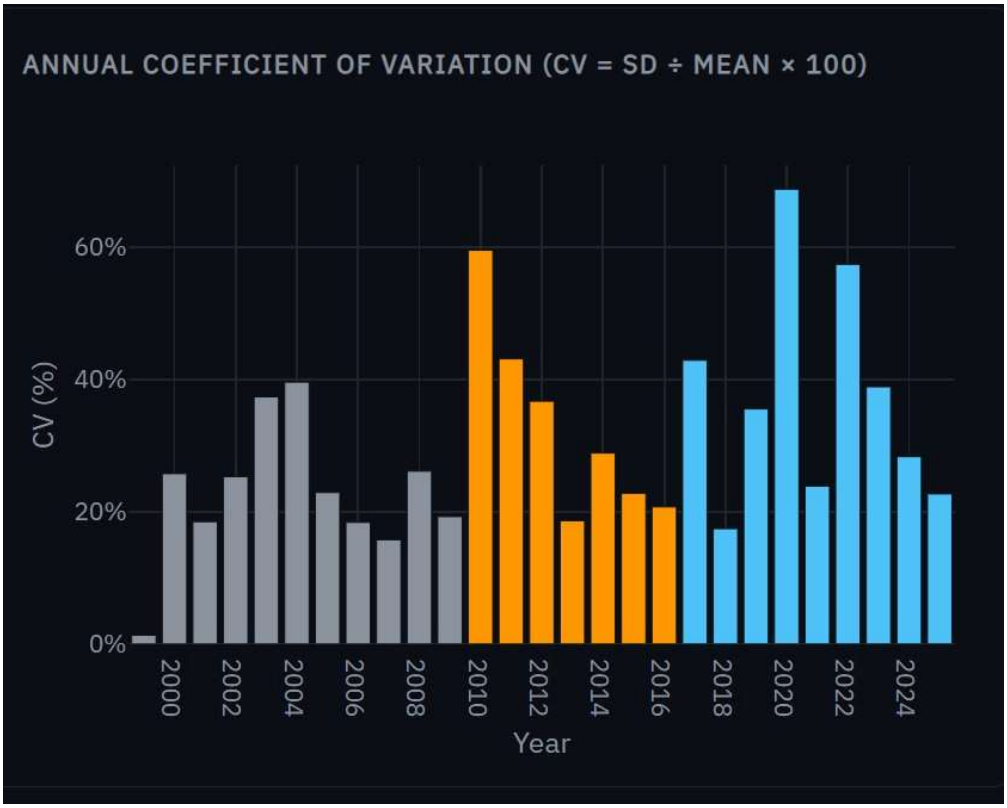

Figure 10. Annual coefficient of variation by era. Grey = pre-renewable (1999–2009), orange = early transition (2010–2016), blue = high renewable (2017–2025). No systematic pattern of higher CV in the high-renewable era.

Visual analysis — emissions intensity co-trend

Figure 11 tracks Australia's emissions intensity (kgCO_2e/MWh) over the full 1999–2025 period. The curve demonstrates that the energy transition is achieving its primary environmental goal: emissions per unit of electricity fell from approximately 1,000 kgCO_2e/MWh in 1999 to roughly 520 kgCO_2e/MWh in 2025 — a 47% reduction. The decline rate accelerates visibly after 2017 (the shaded high-renewable era), coinciding with rapid Wind and Solar deployment. This chart provides important context for the price volatility question: even if prices have become more volatile, the grid is delivering substantially lower-carbon electricity.

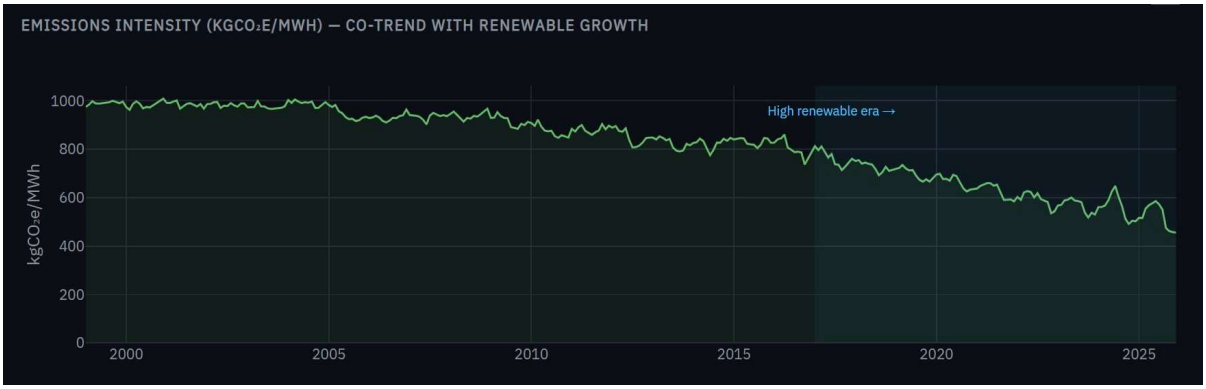

Figure 11. Emissions intensity (kgCO_2e/MWh) for the Australian NEM, 1999–2025. The decline accelerates sharply after 2017, coinciding with the high-renewable era. The shaded region marks the period when Wind+Solar exceeded 10% of generation.

Key findings

Era	Mean price (AUD/MWh)	Std dev	Max price	Avg W+S share
Pre-renewable (1999–2009)	$37.4	$12.7	$69.5	~0%
Early transition (2010–2016)	$47.7	$17.4	$86.3	~5%
High renewable (2017–2025)	**$100.1**	**$52.6**	**$317.1**	~25%

- **Prices rose sharply in the high-renewable era:** Mean wholesale price more than doubled from the early-transition to the high-renewable era ($47.7 → $100.1 AUD/MWh). Standard deviation also increased substantially ($17.4 → $52.6).
- **Weak correlation between W+S share and CV:** The correlation between annual Wind+Solar share and annual CV was r = 0.26, indicating a weak positive relationship. This suggests renewable penetration is not the primary driver of price volatility.
- **2022 as a confounding outlier:** The 2022 peak of $317.1 AUD/MWh — the highest price on record — occurred during the global energy crisis triggered by Russia's invasion of Ukraine and Australian domestic coal supply disruptions. This event, unrelated to renewable penetration, dominates the statistical picture. Excluding 2022 weakens the correlation further.
- **Emissions declining consistently:** Emissions intensity (kgCO_2e/MWh) fell from ~975 in 1999 to ~520 in 2025 — a 47% reduction. This decline accelerates after 2017, coinciding with the rapid deployment of Wind and Solar, confirming that the transition is achieving its primary climate goal regardless of price effects.
- **Price volatility driven by fuel markets, not renewables:** The evidence suggests that fossil fuel market shocks (2007 gas prices, 2017 coal supply tightness, 2022 global crisis) explain most of the observed CV spikes, rather than renewable intermittency. This is consistent with international literature on wholesale electricity market dynamics.

Visualisations used for RQ3

- **Monthly price line chart (with era shading):** Full 1999–2025 price timeline with background shading for the three eras. Allows identification of specific spike events and contextualises when price instability increased.
- **Rolling 12-month SD area chart:** Smoothed volatility signal using a 12-month rolling standard deviation. Better than annual SD for identifying when volatility regimes changed, without being dominated by individual outlier months.
- **Annual CV bar chart (coloured by era):** Enables year-by-year comparison of relative volatility, with era colour encoding providing immediate visual grouping.
- **Scatter plot — W+S share vs CV:** The critical chart for answering RQ3. Each point is one year; trend line with R^2 and slope annotated. An interactive toggle allows exclusion of 2022, demonstrating the sensitivity of the result to the energy crisis outlier.
- **Era box plot:** Price distribution within each era shown as box plots. Median, interquartile range, and outliers visible simultaneously — superior to era-mean tables for showing how the full distribution changed.
- **Dual-axis trend — W+S share and mean price:** Two y-axes showing renewable share growth and price level over time. Makes the co-movement (or lack thereof) visually apparent.

3.4 RQ4 — Energy Portfolio Diversification

Motivation and research objective

RQ1 measured the speed of decarbonisation and RQ2 tested grid resilience during peak stress. However, neither question addresses a deeper structural concern: is the energy transition producing a healthier, more balanced generation portfolio — or merely replacing one dominant fuel with another? A grid that shifts from 70% coal to 70% solar may score well on carbon metrics, but remains structurally fragile. A single technology disruption — prolonged cloud cover, a panel supply-chain failure, or a policy reversal — could destabilise the entire system. RQ4 therefore asks:

> RQ4: Is the energy transition making electricity grids structurally more diverse and resilient, or simply swapping one dominant fuel for another?

The goal is to quantify whether each country's generation mix is becoming more balanced over time, and whether decarbonisation mechanically produces diversification or whether a country can decarbonise without diversifying. This distinction has direct policy relevance: a diversified grid is inherently more resilient to supply shocks, fuel price spikes, and technology-specific disruptions, regardless of its carbon intensity.

Key terminology

This analysis borrows two metrics from competition economics and adapts them to electricity generation:

- **Herfindahl-Hirschman Index (HHI):** A market concentration metric calculated as the sum of squared market shares, scaled to a 0–10,000 range. In its original use, regulators (such as the U.S. Department of Justice) apply HHI to assess whether a merger would create a monopoly. Here, each energy source is treated as a 'firm' and its share of total monthly generation is its 'market share.' An HHI of 10,000 means one source produces 100% of electricity (monopoly). An HHI below 2,500 indicates a diversified mix. Between 2,500 and 4,500 is moderately concentrated. Above 4,500 is highly concentrated. The formula is: $HHI = \sum(share_i^2) \times 10{,}000$, where share_i is the fractional contribution of source i to total generation.
- **Effective Number of Sources (ENS):** An intuitive transformation of HHI defined as ENS = 10,000 ÷ HHI. If a grid has six sources but one dominates at 70%, the ENS might be only 2.1 — meaning the grid behaves as if it has just two equally-sized sources. If all six sources contributed equally (~16.7% each), the ENS would be 6.0. A rising ENS over time indicates genuine structural diversification, not just a token addition of minor sources.
- **Top source dominance:** The share of total generation contributed by the single largest source in each country-year. A declining top-source share confirms that diversification is real — the dominant fuel is losing its grip, not just being joined by marginal contributors.
- **Diversification speed:** The slope of a linear regression fitted to the monthly HHI time series for each country. A more negative slope means faster diversification. This metric is directly analogous to the transition velocity (regression slope on fossil share) computed in RQ1, enabling a consistent cross-country comparison.

Approach and justification

HHI was computed for each country-month from the combined dataset. The same combined_energy.csv used for RQ1 and RQ2 provides the input data, requiring no additional data collection. For each month, the generation_gwh values for all sources present in a country were summed to give the total, then each source's share was squared and summed. The HHI was tracked as a monthly time series with linear regression trend lines overlaid. Annual averages of HHI, ENS, and top-source share were computed for summary tables. A scatter plot of monthly HHI against monthly fossil share was constructed to test whether decarbonisation mechanically produces diversification — if the two are correlated, then reducing fossil dependence simultaneously improves structural resilience.

Visual analysis — diversification speed and top source dominance

Figure 12 presents two panels side by side. The left panel shows the diversification speed — the linear regression slope of monthly HHI for each country. Australia's bar extends to −18.7 HHI points per month, dwarfing the UK (−5.1) and the US (essentially zero). The right panel tracks the share of the single largest source per country over time. Australia's dominant source (Coal) dropped from 66% in 2019 to 50% in 2025 — a 16 percentage point decline in dominance. The UK's dominant source (Gas) declined from 46% to 35%. The US's dominant source (Gas) remained flat at ~40% across the entire window, confirming that its grid structure is frozen despite modest compositional changes.

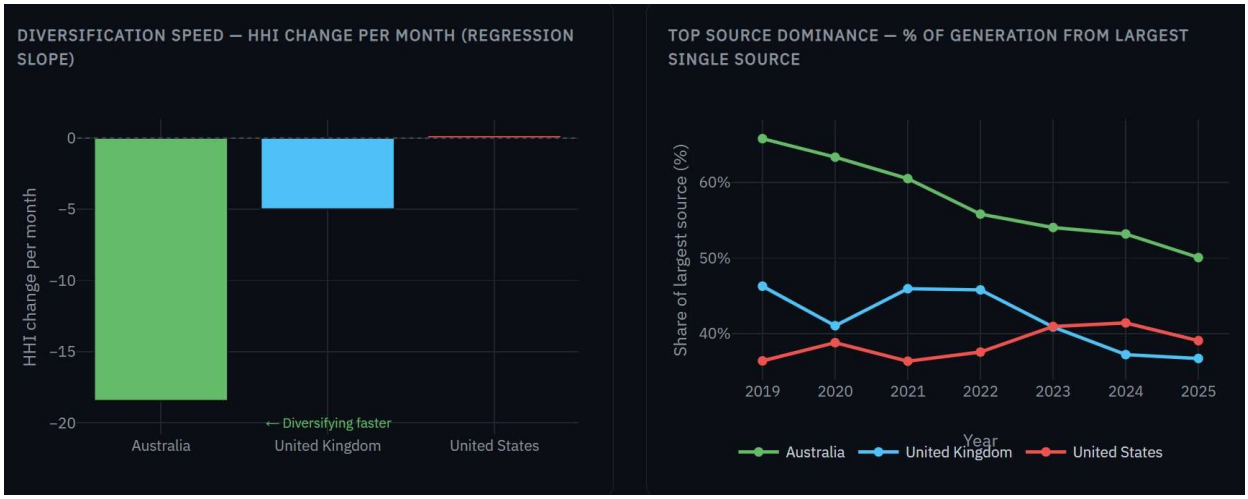

Figure 12. Left: Diversification speed (HHI change per month). Australia diversifies 3.7× faster than the UK and effectively infinitely faster than the static US. Right: Top source dominance declining over time — Australia's Coal dropped from 66% to 50%; UK Gas from 46% to 35%; US Gas stable at ~40%.

Visual analysis — does decarbonisation drive diversification?

Figure 13 is the most analytically important chart in this section. Each dot represents one country-month, plotted by its fossil share (x-axis) against its HHI concentration index (y-axis). Dashed lines show per-country linear regression fits. The pattern is strikingly clear: for all three countries, months with higher fossil share have higher HHI (more concentrated), and months with lower fossil share have lower HHI (more diverse). This establishes a direct structural link between decarbonisation and diversification.

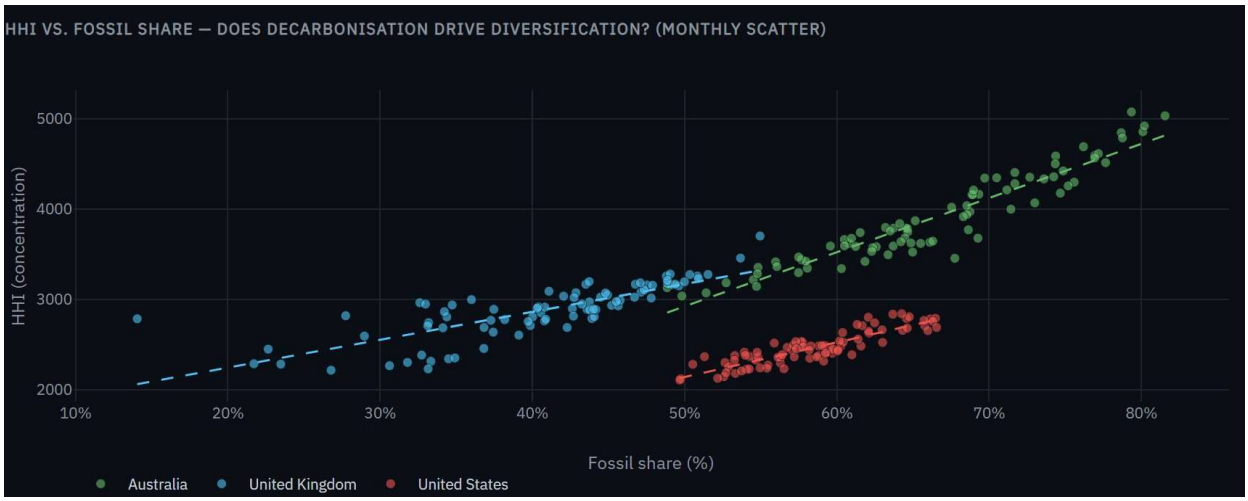

Figure 13. HHI vs. fossil share — monthly scatter with per-country trend lines. Strong positive correlation across all three countries confirms that reducing fossil dependence mechanically increases portfolio diversity. Australia (green) spans the widest range, reflecting its rapid transition from 80% to 50% fossil.

Three observations emerge from Figure 13. First, Australia (green) spans the widest x-axis range (50–80% fossil), reflecting the magnitude of its transition. Its trend line has the steepest slope, meaning each percentage point of fossil reduction yields a larger diversity gain than in the other two countries — because Australia is displacing a single dominant source (Coal) with multiple smaller ones (Solar + Wind + Gas). Second, the UK (blue) occupies a lower and narrower band (15–55% fossil), with a similarly strong positive slope. Its diversity improvements come from Gas declining while Wind and Nuclear maintain stable shares. Third, the US (red) clusters in a tight blob around 55–65% fossil and HHI 2,000–2,900 — confirming that the US grid is structurally stagnant. Its trend line is nearly flat, meaning that the modest fossil share decline it has achieved is not translating into meaningful diversity improvements.

Key findings

Country	HHI 2019	HHI 2025	Δ HHI	Slope (pts/mo)	ENS 2019	ENS 2025	Dominant source
Australia	4,650	**3,362**	**−1,288**	**−18.7**	2.2	**3.0**	Coal (66% → 50%)
United Kingdom	3,071	2,627	−444	−5.1	3.3	3.8	Gas (46% → 35%)
United States	2,481	2,391	**−90**	**+0.2**	4.0	4.2	Gas (stable ~40%)

- **Australia diversifying fastest:** Australia's HHI dropped by 1,288 points (from 4,650 to 3,362), moving from 'highly concentrated' to 'moderately concentrated' territory. Its ENS rose from 2.2 to 3.0 — the grid evolved from a de facto two-source system (Coal + everything else) to a genuine three-source mix (Coal + Solar + Wind). The HHI slope of −18.7 points per month is 3.7× faster than the UK's diversification rate.
- **UK steadily broadening:** The UK entered 2019 with a moderately concentrated grid (HHI = 3,071) dominated by Gas at 46%. By 2025, Gas declined to ~35% while Wind rose to match it, producing a more balanced portfolio with HHI = 2,627 and ENS = 3.8. The UK is approaching the 'diversified' threshold of HHI < 2,500.
- **US already diverse but structurally frozen:** The US started as the most diverse grid (HHI = 2,481, ENS = 4.0) thanks to its established mix of Gas, Coal, Nuclear, and Wind.

However, its HHI barely changed (slope = +0.2 pts/month), meaning the transition is not making the US grid more or less diverse — it is structurally frozen. Gas remains dominant at ~40% across the entire window. This is a paradoxical finding: the US is the most diverse but the least dynamic.

- **Decarbonisation drives diversification:** The scatter plot (Figure 13) demonstrates a strong positive correlation between fossil share and HHI across all three countries. The displacement of a dominant fossil fuel mechanically increases diversity, because the replacing sources (Wind + Solar) enter at smaller individual shares. This finding has a policy implication: investing in renewable deployment achieves two goals simultaneously — lower carbon intensity and higher structural resilience.

Visualisations used for RQ4

- **Diversification speed bar chart (Figure 12, left):** HHI regression slope per country. Negative = diversifying. The bar chart idiom was chosen for the same reason as the RQ1 transition speed chart (Figure 5) — it enables direct visual comparison of a single scalar metric across countries. The zero reference line and directional annotation ('← Diversifying faster') guide interpretation.
- **Top source dominance line chart (Figure 12, right):** Annual share of the largest single source per country. Tracks whether the dominant fuel is losing its grip over time. A line chart was chosen because the temporal trend is the key signal — a declining line confirms genuine structural change, not just statistical noise.
- **HHI vs. fossil share scatter (Figure 13):** The critical analytical chart for RQ4. Monthly data points coloured by country with per-country linear trend lines. The scatter plot idiom was chosen because it reveals the relationship between two continuous variables — a question that line or bar charts cannot address. Each country's cluster position, spread, and slope tell a distinct story about its structural trajectory.
- **Monthly HHI line chart (in dashboard):** Time series with background bands showing concentration thresholds (green < 2,500, amber 2,500–4,500, red > 4,500). Dashed trend lines overlay raw monthly data. The colour-coded bands provide immediate context for whether a country's HHI level is concerning or healthy.
- **ENS line chart (in dashboard):** The same data transformed via ENS = 10,000 ÷ HHI. More intuitive than raw HHI — readers immediately understand '3.0 effective sources' without needing to learn the HHI scale.
- **Stacked bar — annual source shares (in dashboard):** Full generation mix faceted by country, enabling visual inspection of which specific sources are gaining or losing share. Confirms that diversification is driven by Wind and Solar entering the mix, not by shuffling among existing fossil sources.

4. Conclusion

This project set out to answer four questions about the global energy transition. The data provided clear answers to three and a nuanced answer to the fourth.

RQ1 is answered decisively: Australia transitioned fastest, recording a fossil share decline of 0.289 percentage points per month and a Wind+Solar gain of 0.298 pp/month from 2019 to 2025 — approximately 8× faster than the US on fossil decline (Figure 5). The UK maintained a steady transition from an already-lower fossil base. The US moved slowest, reflecting structural and political barriers that short analysis windows cannot overcome.

RQ2 revealed that all three grids remain fossil-dependent during peak stress, but to varying degrees. The UK is most resilient (reliability gap of just 1.9 pp, Figure 7), supported by Nuclear as firm baseload. The US is most vulnerable (gap of 4.1 pp), with Gas and Coal covering 63.4% of peak generation (Figure 6). Australia's backup mix remains coal-dominated despite rapid renewable growth, though year-on-year improvement is measurable (Figure 8).

RQ3 produced the most nuanced finding: the correlation between Wind+Solar penetration and price volatility in Australia is weak ($r = 0.26$), and the dominant volatility events are attributable to fossil fuel market shocks rather than renewable intermittency. The 2022 energy crisis provides a stark natural experiment — price volatility spiked to its highest level precisely when global gas and coal markets were most stressed, not when renewables were highest.

RQ4 revealed an important structural benefit of the transition that RQ1 alone could not capture: Australia's grid is not merely decarbonising — it is diversifying. Its HHI fell by 1,288 points (the equivalent of gaining one full effective source), transforming a coal-dominated monoculture into a genuine multi-source grid. The UK followed a similar pattern at a slower pace. The US, despite being the most diverse grid in absolute terms, is structurally frozen — its diversity is not improving. This finding has direct policy implications: a diversified grid is inherently more resilient to supply shocks, fuel price spikes, and technology-specific disruptions.

5. Reflection

The most important lesson from this project is that data wrangling consumes the majority of effort and that datasets from official sources are far from clean. The readr type coercion bug in the US dataset, the 4 million duplicate rows, the 2,888 sensor dropout readings in GridWatch, and the leading whitespace on every UK timestamp were all invisible until explicitly investigated. A key principle that emerged: never trust that a dataset is clean because it comes from a government agency — always verify programmatically.

On visualisation, the decision to use interactive Shiny dashboards rather than static figures was vindicated by the exploration process: the ability to filter by country, adjust year ranges, and toggle trend lines repeatedly revealed patterns that would have been missed in fixed charts. The addition of RQ4 and the HHI metric demonstrated the value of applying analytical tools from adjacent fields

— economics in this case — to energy data. In hindsight, a greater investment in annotation — labelling specific events (2022 energy crisis, the UK coal closure date, Australia's policy inflection points) directly on the charts — would have made the findings more self-explanatory to a general audience.

A limitation of the project is the absence of price data for the UK and US, which constrains RQ3 to Australia alone. Future work could incorporate UK Balancing Mechanism data from Elexon and US day-ahead market prices from CAISO or PJM to enable a genuine cross-country price volatility comparison. Additionally, sub-monthly granularity (hourly or daily prices) would allow a much more rigorous test of the renewable-volatility hypothesis, as monthly averages smooth out the intraday price swings that drive retailer and consumer risk.

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